

Data Converters

Exercise 02 – INL, SFDR, DACs, and Sampler

Task I – INL & SFDR

The power spectral density plot of an 11-bit ADC for a full-scale sinusoidal input signal is shown in Figure 1.

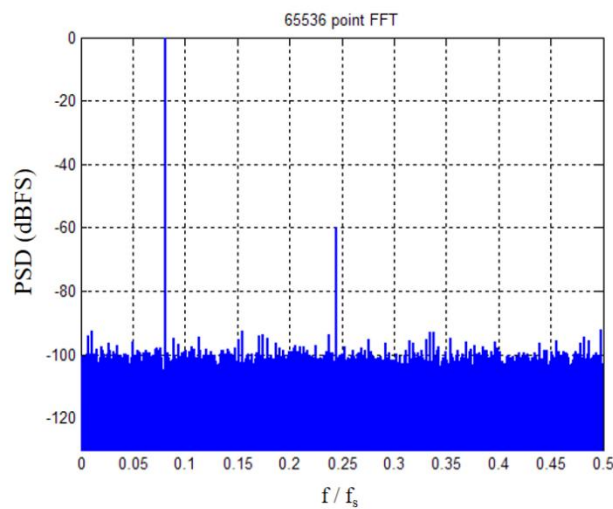


Figure 1: PSD plot of an 11-bit ADC.

- a. Determine the SFDR.

$$\text{SFDR} = 60 \text{ dBFS}$$

- b. Consider the spectrum and determine a_1 and a_3 in terms of FS .

Note: The non-linear transfer function of the ADC can be approximated by

$$\text{out}(v_{\text{in}}) = a_1 v_{\text{in}} + a_3 v_{\text{in}}^3$$

since

$$\sin^3 x = \frac{3}{4} \sin x - \frac{1}{4} \sin(3x)$$

$$a_1 = 0.997, a_3 = \frac{0.004}{FS^2} = \frac{0.016}{FSR^2}$$

- c. Determine the slope of the ideal transfer characteristic in terms of a_1 , a_3 , and FS .

$$f_{\text{IDEAL}}(x) = (a_1 + a_3 FS^2) \cdot x$$

- d. Determine the maximum INL of the ADC in $LSBs$.

$$INL_{MAX} \approx 1.58 \text{ LSB}$$

e. Determine the standard deviation of the electronic noise P_{EN} in $LSBs$.

$$\sigma_{EN} \approx 1.28 \text{ LSB}$$

f. Determine the effective number of bits of the ADC.

$$ENOB \approx 8.64 \text{ bits}$$

g. Consider the data sheet of the ADC *LTC2241-12*, in which the INL-plot under “Typical Performance Characteristics” shows a cubic nonlinearity.

- Use your result of task I.d to estimate the *SFDR* of the converter from INL_{MAX} according to the data sheet.

$$SFDR \approx 80 \text{ dB}$$

- Does your result match the typical *SFDR* stated for 10 MHz in the data sheet?

The mathematical result of $SFDR = 80 \text{ dB}$ fits well with the 78 dB stated in the data sheet.

Task II – Redesign of a Segmented DAC

Consider the paper *C.-H. Lin and K. Bult, “A 10-b, 500-MSample/s CMOS DAC in 0.6 mm²,” IEEE J. Solid-State Circuits, vol. 33, no. 12, pp. 1948-1958, Dec 1998*. In this paper, it was found that, for the given objectives, a thermometer/ binary partition of 8/2 is most suitable.

Adapt the number of bits B_t of the thermometer section in order to re-optimize the DAC according to the following specifications:

parameter	description	value
B	resolution	10 bits
INL_{spec}	worst-case integral nonlinearity	< 0.4 LSB
DNL_{spec}	worst-case differential nonlinearity	< 0.2 LSB
Y	target yield	95%
A_{decode}	approximate decoder & routing area	$2^{B_t} \cdot 3000 \mu m^2$
A_{total}	total DAC area (to be minimized)	$2^B \cdot A_{unit} + A_{decode}$
k_u	unit element matching parameter	6 % μm
σ_u	standard deviation of unit element mismatch	$k_u / \sqrt{A_{unit}}$

Note: The inverse of the error function of 0.95 results in 1.96.

Select $B_t = 4$ for minimum total area.

Task III – DNL of a Segmented DAC

Please turn the page.

A segmented DAC consists of a unit-element MSB DAC with B_1 bits. The standard deviation of the unit elements is σ_1 . The number of bits in the binary weighted LSB DAC is B_2 and the standard deviation of the elements is σ_2 .

State an expression for the standard deviation of the DNL for the worst-case code.

$$\sigma_{\text{DNL}}^2 \approx 2^{B_2} I_0 (\sigma_1^2 + \sigma_2^2)$$

Task IV – Sampler

An n-MOS sampler used in the front-end of a B-bit ADC is shown in Figure 2.

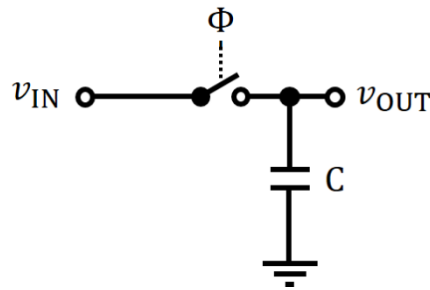


Figure 2: Sampler.

The RC time constant of this circuit was adapted such that the sampler settles to within one LSB in one half clock period.

Derive an expression for the total thermal noise of the sampler in terms of the resistance R , the Boltzmann constant B , the sampling frequency f_s , and the temperature T .

$$P_{\text{th}} = \frac{kT}{C} = 2kTf_s R \ln 2^B$$